

On really big numbers

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History of writing numbers

How big numbers people needed in:

- ▶ prehistoric times
- ▶ beginning of agriculture
- ▶ first cities
- ▶ empires
- ▶ modern times - let's discuss that later

History of writing numbers - Babylon

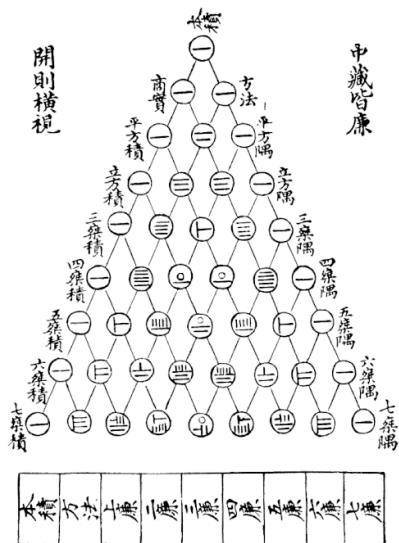
 1	 11	 21	 31	 41	 51
 2	 12	 22	 32	 42	 52
 3	 13	 23	 33	 43	 53
 4	 14	 24	 34	 44	 54
 5	 15	 25	 35	 45	 55
 6	 16	 26	 36	 46	 56
 7	 17	 27	 37	 47	 57
 8	 18	 28	 38	 48	 58
 9	 19	 29	 39	 49	 59
 10	 20	 30	 40	 50	

History of writing numbers - Maya

0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
—	• —	•• —	••• —	•••• —
10	11	12	13	14
==	• ==	•• ==	••• ==	•••• ==
15	16	17	18	19
===	• ===	•• ===	••• ===	•••• ===

History of writing numbers - China

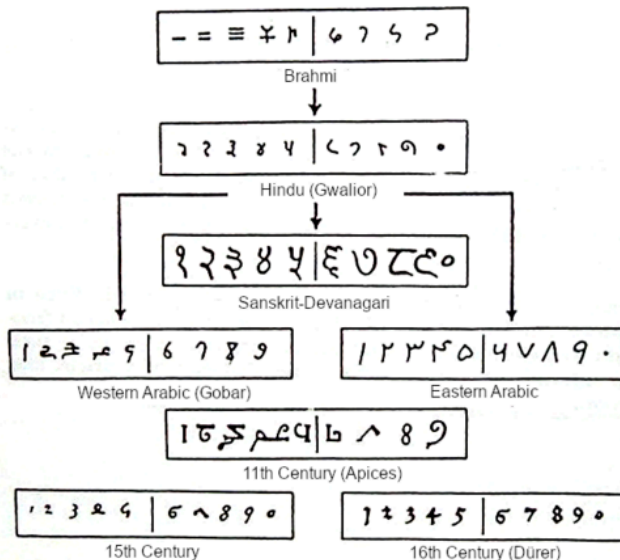
古法七葉方圖



History of writing numbers - Roman Empire



History of writing numbers - Indian → Arabic



History of writing numbers - Modern times

Positional decimal system

- ▶ $2026 = 2 \times 10^3 + 0 \times 10^2 + 2 \times 10^1 + 6 \times 10^0$.
- ▶ Powers - good tricks!
- ▶ $3.7 \times 10^{13} = 37,000,000,000,000$ – big number
- ▶ $2 \times 10^{-9} = 0.000000002$ – small number

How big/small numbers we need?

- ▶ **Economy:**

trillions – $1,000,000,000,000 = 10^{12}$

US \$1 trillion, 1 billion GBP

- ▶ **Technology:**

quadrillions – $1,000,000,000,000,000 = 10^{15}$

1PB = 1 Petabyte.

- ▶ **Biology:**

$3.7 \times 10^{13} = 37000000000000$ – approximate number of cells in a human body

$8.6 \times 10^{10} = 86000000000$ – number of neurons in a human brain

6.4×10^9 – base pairs in a DNA

$7\mu\text{m} = 7 \times 10^{-6}\text{m}$ - the size of a cell nucleus

How big/small numbers we need?

► Physics:

10^{12} galaxies in the Universe

10^{23} stars in the Universe

10^{80} atoms in the Universe

8.8×10^{26} m – the diameter of the Universe

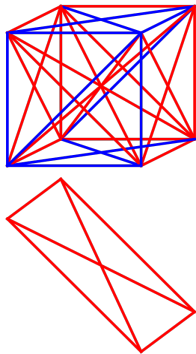
$1.6 \text{ fm} = 1.6 \times 10^{-15} \text{ m}$ – the size of a neutron

$< 10^{-19} \text{ m}$ – the size of an electron

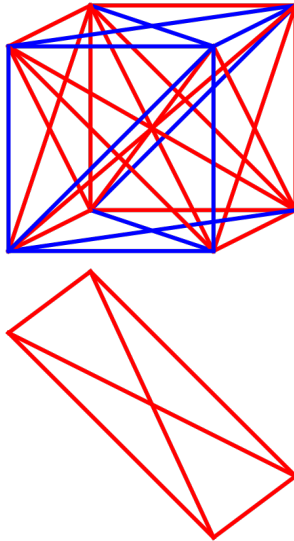
Conclusion: The positional decimal system is sufficient!

How big numbers we need in **Maths**?

Problem: Connect each pair of geometric vertices of an n -dimensional hypercube to obtain a complete graph on 2^n vertices. Colour each of the edges of this graph either red or blue. What is the smallest number n for which every such colouring contains at least one single-coloured complete subgraph on four coplanar vertices?



How big numbers we need in **Maths**?



Solution: Ron Graham (1935-2020)



Graham showed that such number n exists and

$$6 \leq n \leq G,$$

where G is *Graham's number*. So what is Graham's number?

Knuth's notation

- ▶ $a^b = a \uparrow b$
- ▶ $2 \uparrow 4 = 2 \times (2 \times (2 \times 2)) = 2^4 = 16$
- ▶ $2 \uparrow\uparrow 4 = 2 \uparrow (2 \uparrow (2 \uparrow 2)) = 2^{2^{2^2}} = 2^{16} = 65,536$
- ▶ $2 \uparrow\uparrow\uparrow 4 = 2 \uparrow\uparrow (2 \uparrow\uparrow (2 \uparrow\uparrow 2)) = 2^{2^{\cdot^{\cdot^{\cdot^2}}}}$ – tower of 65,536 2's
= $2 \uparrow^3 4 = 2 \uparrow^2 (2 \uparrow^2 (2 \uparrow^2 2))$ – simpler notation
- ▶ $2 \uparrow^n 4 = 2 \uparrow^{n-1} (2 \uparrow^{n-1} (2 \uparrow^{n-1} 2))$ – madness!

Graham's number

- ▶ $G_1 = 3 \uparrow^4 3 = 3 \uparrow\uparrow\uparrow\uparrow 3$ – tower of trillions of 3's
- ▶ $G_2 = 3 \uparrow^{G_1} 3$
- ▶ $G_3 = 3 \uparrow^{G_2} 3$
- ▶ $\vdots$
- ▶ $G_{64} = 3 \uparrow^{G_{63}} 3$

GRAHAM'S NUMBER

$$G = G_{64} = 3 \uparrow^{G_{63}} 3.$$

Graham's number



Graham's number is so big that if we wanted to write its decimal expression in digits as small as electron, then the Universe would be too small to contain it. Worse! We know that it is a tower

$3^{3^{\dots^3}}$, but even this tower is too big for the Universe! On the other hand, we know that the last 10 digits of G are

...2464195387.

Any questions?

Further reading



Thank you!